

Automatic Fault Detection And Analysis Of Electric Vehicle Charging Station By Machine Learning

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Abstract— As the transition to Electric Vehicles (EVs) accelerates, enhancing the resilience of electrical systems becomes increasingly crucial. A key component in EVs is the Power Electronic Converter (PEC), which not only powers the electric motors but also reconfigures the grid charging of the battery bank. This paper introduces a fault detection system specifically designed for a prevalent type of charger used in EVs. The system aims to identify faults that may lead to malfunctions during the operation of the onboard electric charger, thereby enabling a supervisory system to execute corrective measures and maintain uninterrupted charging operations. The proposed fault detection method leverages a Machine Learning (ML) technique known as a classification algorithm. The system is trained using grid-side Total Harmonic Distortion (THD) and load resistance values to classify potential faults accurately. Extensive testing demonstrates that the proposed system achieves a fault prediction accuracy of around 94.59 %, showcasing superior performance in terms of training and execution times as well as overall accuracy. This study underscores the effectiveness of integrating advanced ML techniques in enhancing the reliability and efficiency of EV charging systems, thus contributing to a smoother and faster transition to electric mobility.

Keywords— machine learning, EV, charging station, fault detection, total harmonic distortion, power electronic converter.

I. INTRODUCTION

The proliferation of EVs has highlighted the critical need for robust charging infrastructure. These systems enable fast charging by connecting vehicles to their power sources. Given the relatively recent adoption of this technology, diagnosing and repairing faults in charging devices is often challenging and time-consuming. Therefore, developing a comprehensive theoretical framework and conducting thorough technical research in this field is essential [1]. If a power battery fails during charging, it could result in a fire accident, leading to significant economic losses. Thus, predicting battery failures, issuing early warnings, and implementing preventive measures are crucial during EV charging to mitigate such risks [2], as seen in Fig. 1, which depicts the impacts of EV charger faults. A residual mechanism applied fault detection and isolation (FDI) was examined in battery charging equipment. It isolated multiple faults by generating distinct residual

signals. Various faults, such as battery short/open circuits, sensor biases, input voltage drops, and switch faults, were detected and isolated using the magnitude, slope, or norm of residual signals from battery voltage [3]. A data-driven method for diagnosing battery charging capacity used real-world EV data, including charging rate, temperature, state of charge (SOC), and mileage [4]. A tree-based prediction model with polynomial features was developed, and a statistics-based method analyzed error distributions to identify abnormalities [5].

The work reported in [6] aimed to identify faults in the onboard converter using the Random Forest Classifier (RFC), focusing on semiconductor components and grid status during battery charging. It compared the performance of RFC system and Support Vector Machine (SVM) classifies system, showing that RFC had better training and execution times, accuracy, and generalizability to other faults. A data-driven, human-interpretable method [7] was presented for detecting anomalies in EV charging stations. The approach aims to enhance condition monitoring and predictive maintenance of the power converters in these stations. A fault-tree based analysis method [8] was applied to an expert system via a database. Simulations showed that the system met the requirements for DC charging infrastructure fault diagnosis. Researchers have categorized the primary faults of the electric drive system according to its self-test and online diagnostic needs [9]. They introduced both a self-test fault tree and an online diagnosis fault tree for the system. In [10], authors demonstrated a new fault diagnosis approach for EV battery systems, leveraging big data statistical methods. Utilizing neural network algorithms, it integrated fault and defect diagnosis outcomes with statistical regulations on big data to develop a comprehensive battery system fault diagnosis model.

Accurate and timely fault diagnosis for EV charging equipment holds significant importance [11]. Using semi-supervised learning, the fault characteristics of charging equipment are utilized for classifying and identifying charging equipment faults. Experimental results demonstrate that this method significantly enhances the accuracy of charging equipment fault classification, achieving over 90% accuracy

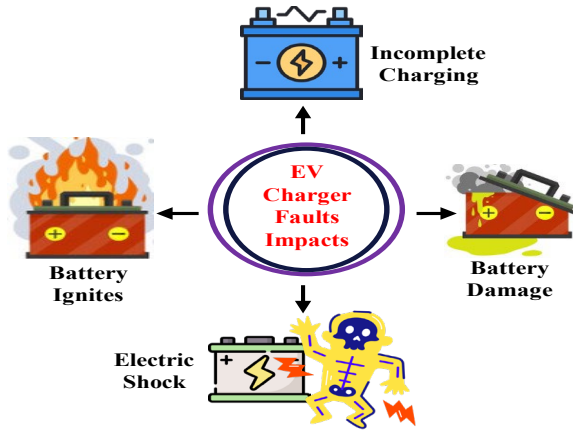


Fig. 1. Various impacts of faulty EV charger.

across all fault types. To address the inadequacies of traditional thresholding methods in identifying anomalies [12] during EV charging, a new method based on a multivariate Gaussian distribution model is proposed in [13]. It involves preprocessing data, visualizing anomaly categories and standard curves, and using the model to predict anomalies in new data. An effective abnormal prediction was proposed in [14] for EV charging equipment by analyzing extensive operational data and employing various ML techniques. The XGBoost algorithm stood out with a 90.34% accuracy rate, showcasing its superiority in predicting abnormalities in equipment components and characteristics.

Accurately predicting faults in EV chargers is essential for anticipating battery failures [15], issuing timely warnings [16], and taking preventive actions to reduce risks during charging [17]. While existing research has made valuable contributions, significant gaps remain in the inefficient prediction of anomalies in EVs using ML, particularly in terms of achieving high accuracy rates. This research contributes a novel fault detection method for onboard electric chargers, utilizing a ML-based classification algorithm. With proposed method offers a very high fault prediction accuracy because of integrating advanced ML techniques to enhance reliability and efficiency in EV charging systems. This advancement accelerates the transition to electric mobility by ensuring uninterrupted charging operations with superior accuracy.

II. PROPOSED SYSTEM CONFIGURATION

A. System Overview

The system for analyzing faults in an electric vehicle (EV) charger consists of three main subsystems, namely grid, converter, and control unit, as shown in Fig. 2. These components work together to simulate and monitor the performance and reliability of the EV charging system under various conditions. The parameters to be considered in the EV charging system are outlined in Table I. The grid model represents the source of electrical power. It simulates the characteristics of the incoming electricity, providing vital inputs, such as current, phase, and voltage data to the control unit. This subsystem ensures that the simulated grid behaves realistically, capturing the dynamics and variations that might be encountered in real-world scenarios. The converter subsystem transforms the alternating current (AC) from the grid into a direct current (DC) suitable for charging an EV. It interfaces with both the grid and the control unit, receiving control signals to modulate its operation. This subsystem is

crucial for maintaining efficient and stable power conversion, essential for effective EV charging. The control unit is at the heart of the proposed system, which monitors and balances the overall operation. It receives input data from the grid, processes this information, and sends appropriate control signals to the converter. Additionally, the control unit monitors faults by receiving fault signals from a dedicated fault detection block. This capability allows the system to promptly identify and respond to issues, ensuring safe and reliable operation. The control unit also references an external data file containing parameters or data essential for accurate simulation and fault analysis. The load subsystem, represented by a resistor block, signifies the system's output where the converted DC power is delivered. This element models the end-use scenario of the EV charger, providing a realistic load that the system must supply. The interplay between these subsystems and the integration of fault detection mechanisms requires a comprehensive ML approach, which is taken in this model to ensure the robustness and effectiveness of EV charging infrastructure.

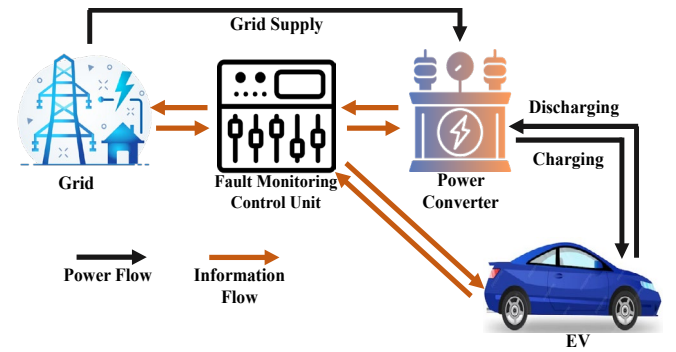


Fig. 2. System architecture of proposed model.

TABLE I. FAULT DETECTION SYSTEM PARAMETERS

Parameters	Value
Grid Frequency	50 Hz
Grid Voltage	415 V
Rectifier Voltage	800 V
IGBT Threshold Voltage	3 V
IGBT On- State Resistance	0.001 Ohm
PWM Switching Frequency	10,000 Hz
Load Resistance	11.2 Ohm

B. Control Scheme of Different Subsystems

An ideal voltage source is a theoretical circuit element that provides a constant voltage regardless of the current drawn from it. Besides, grid subsystem contains a sine wave, signifying an AC source. This block represents an ideal voltage source, which maintains a constant voltage regardless of the load it supplies. Another block within the grid subsystem represents the frequency of the AC voltage. The control unit block receives feedback from the converter block, which could be in the form of voltage or current signals. The control unit uses this feedback to regulate the converter's operation, possibly by adjusting a switch or control signal represented by another arrow returning to the converter block. This feedback loop ensures that the converter delivers the desired voltage and current to the load, typically a battery. The load block, represented by a simple resistor, acts as an electrical load on the converter, mimicking the battery being charged. While an actual battery might be more complex, using a resistor simplifies the model for this analysis, allowing a straightforward representation of the load on the system.

The converter subsystem is depicted in Fig. 3 as a three-phase with a two-level converter utilizing an Insulated-Gate Bipolar Transistor (IGBT), one type of ideal switching circuit. The control unit subsystem manages this three-phase converter through $i_d - i_q$ control, ensuring unity power-factor operation. Each of the three phases consists of an IGBT and a diode connected in series. The control unit controls these IGBTs with switching signals. The DC terminals of the three phases are connected at a point. A capacitor is connected across the positive and negative output terminals of the converter. This capacitor helps to smooth out the DC voltage output. The DC outputs from the three phases are combined at a positive output terminal and a negative output terminal. A capacitor smooths the DC voltage output. The AC side of the converter is connected to the grid via three inductors. These inductors help to limit the rate of change of current on the AC side of the converter. The inductors are connected to the grid via three ideal voltage sources that represent the three phases of the AC grid supply. The Control Unit uses a method called $i_d - i_q$ control to regulate the IGBTs. The proposed control method ensures a constant 800 V_{DC} output voltage while maintaining efficient power drawn from the grid.

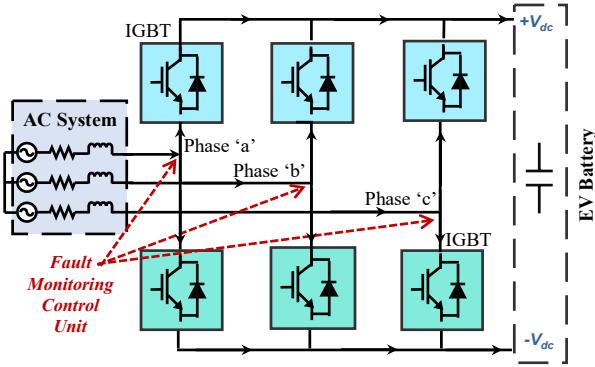


Fig. 3. Control scheme of converter subsystems.

C. Fault Detection

To propose an onboard charger efficient approach targeting failures, it is essential to first define the fault classes. In this paper, a fault is an abnormal condition in the gate drivers of a charger, identified through changes in Total Harmonic Distortion (THD) and variations in load resistance. These faults are categorized as occurring in phase 'a', phase 'b', phase 'c', or combinations of these phases. THD measures the distortion in the electrical signal, which can indicate faults by showing deviations from normal operation. By varying load resistance values, different fault scenarios are simulated to observe how the system behaves under these conditions. This comprehensive approach ensures that various possible states of the system, including normal and faulty conditions, are covered. The faults are categorized and classified to distinguish between normal operation and specific fault conditions in different phases.

III. CLASSIFICATION OF ALGORITHM

A. Ensemble Bagged Tree Algorithm

Ensemble bagging is one type of Machine Learning (ML) technique that enhances accuracy and reduces variance by combining multiple models, as illustrated in Fig. 4. This approach encompasses training several models on various subsets of the training data followed by the combination of the prediction to reach the final decision [18]. Tree-based

ensemble bagging offers significant advantages by reducing overfitting and ameliorating the resilience of models, leading to better performance on unseen data. This approach is adept at managing complicated feature interpretations, which is common in practical scenarios [19]. Widely adopted in both classification and regression tasks, tree-based ensemble bagging is a popular ML technique. It has found applications across numerous sectors, such as smart grid, energy, electrical power, finance, healthcare, and fraud detection. Overall, this method effectively improves the accuracy and generalizability of decision tree models, making it an efficient strategy for machine learning professionals aiming to boost their model's performance.

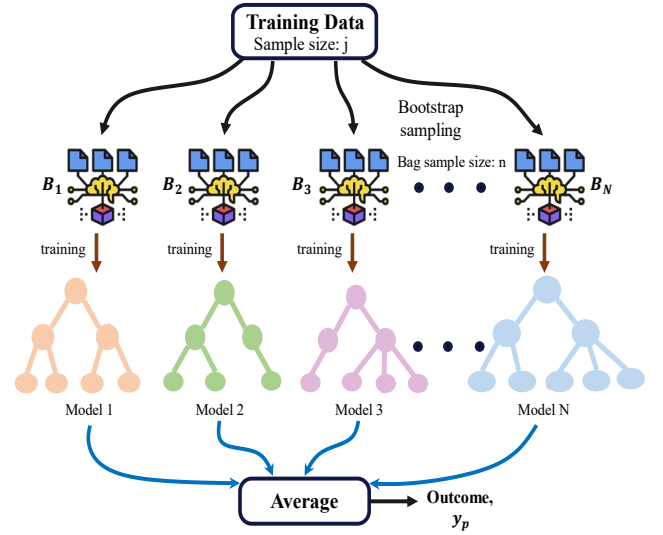


Fig. 4. Schematics of the proposed ensemble bagged tree algorithm.

The steps of ensemble bagging model is presented in Alg. 1 [18]:

Algorithm 1: Ensemble bagging algorithm

1. Input: Training dataset D with n samples, number of models N , number of attributes m and number of bootstrap samples p .
2. Output: Final ensemble model $H(x)$.
3. Initialize: The final ensemble model $H(x) = \{\}$.
4. For $T = 1$ to N do
 1. Draw a bootstrap sample D_T of size p from D .
 2. Select q attributes from the m available attributes.
 3. Train a prediction model $h_T(x)$ using the dataset D_T and the selected q attributes.
 4. Add $h_T(x)$ to the ensemble $H(x)$
5. End For
6. For each input sample x do
 1. For classification problems:

Compute the majority vote or average class score all models $h_T(x)$.
 2. For numerical prediction problems

Compute the average prediction value across all models $h_T(x)$.
7. End For

B. Fault Detection and Monitoring Process

Multiple instances of fault behavior were simulated by varying load conditions and introducing different fault scenarios, creating a comprehensive dataset that covered various possible states of the system. Both fault and no-fault conditions were simulated to ensure that the classifier could distinguish between normal operation and fault conditions. For training data preparation, the THD on the grid side was used as one of the features for classification, as THD measures the distortion in the electrical signal, which can indicate faults. Different load resistance values were used to simulate how the system behaved under varying load conditions, helping the classifier learn how these variations affected the presence of fault. Algorithm 2 provides the fault detection procedure step by step of proposed model. Fig. 5 outlines the working flow of the diagram for fault prediction of proposed system.

Algorithm 2: Fault Detection Method

1. Initialize grid voltage and load options
2. Load and preprocess training data
3. Convert gate driver fault to training data
4. Train classifier by using THD and load value
5. Load and preprocess test data
6. Convert gate driver fault to categorical
7. Predict faults on test data
8. Save trained classifier
9. Initialize simulation parameters
10. For each gate driver fault, do
 - Calculate real faults
11. For each phase, do
 - Extract THD data in time range
12. Prepare measured data matrix
13. Predict faults
 - If predicted fault == real fault, then
 - Display "Prediction: Correct!!"
 - Else
 - Display "Prediction: Incorrect!!"

The synthetic data generated from the simulations, including grid-side THD and load resistance values, was collected and used as training data. The features selected for training included system load and grid THD values for each phase 'a', 'b', and 'c'. The target variable for classification was the fault status of the gate drivers, categorized into different classes such as no fault, fault in phase 'a', fault in phase 'b', fault in phase 'c', and their combinations. The classification algorithm used was a Bagged Trees Ensemble, which aggregated multiple decision trees to improve classification accuracy and stability. The ensemble model was trained on the collected data, learning to associate different patterns in the features (THD and load resistance) with specific fault conditions. To ensure the model's robustness and to prevent overfitting, cross-validation was performed. The dataset was split into folds, and the model was trained and validated on different subsets of the data. Validation accuracy was computed to assess the effectiveness of the model in correctly identifying the fault conditions. This approach ensured that the classifier was well-equipped to detect and classify faults in different phases of the electrical system, contributing to more efficient fault diagnosis and maintenance.

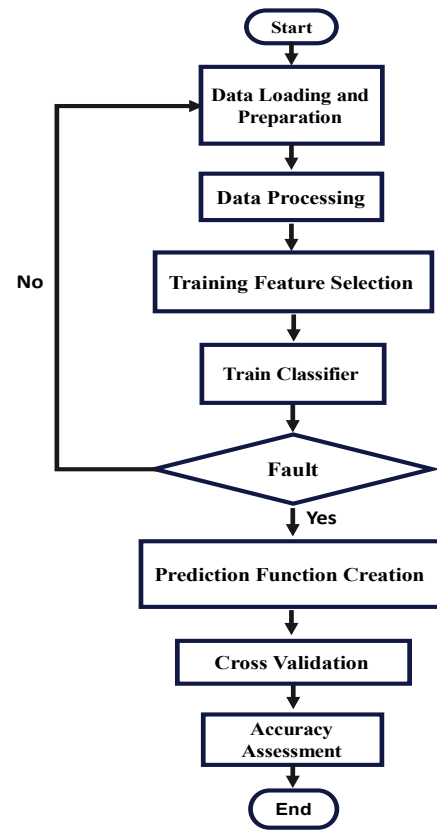


Fig. 5. Working flow of proposed fault prediction approach.

IV. PERFORMANCE EVALUATION AND DISCUSSION

Fig. 6 depicts grid voltage and current over time. The top graph illustrates the voltage waveforms for a three-phase system, showing the sinusoidal nature of the voltages, which are out of phase with each other by 120 degrees. This indicates a balanced three-phase system, essential for efficient power distribution and conversion. The bottom graph depicts the corresponding current waveforms for the same three-phase system. The currents also exhibit sinusoidal behavior but show some variations and distortions, suggesting dynamic interactions within the system or the presence of harmonics. The phase currents are similarly out of phase by 120 degrees, maintaining the balance crucial for system stability.

The current waveforms seem to have slight deviations from the ideal sinusoidal shape, possibly due to the load characteristics or converter switching effects. Both graphs together provide a comprehensive view of the power flow and quality in the grid-connected system. The consistent pattern in voltage and current indicates effective control and conversion mechanisms, with the observed distortions offering insights into potential areas for optimization or further investigation. Besides, Fig. 7 provides a detailed analysis of inductor current, power dissipation, and voltage over time. The top graph displays the current through an inductor, showing a generally steady trend with minor fluctuations. This indicates that the inductor is functioning as expected, smoothing out current variations and maintaining stability within the circuit. The second graph illustrates the power dissipation in the system. The power dissipation exhibits periodic peaks and troughs, suggesting that the system is experiencing varying loads or operational conditions. These fluctuations are due to changes in the circuit's demand or the switching behavior of the converter.

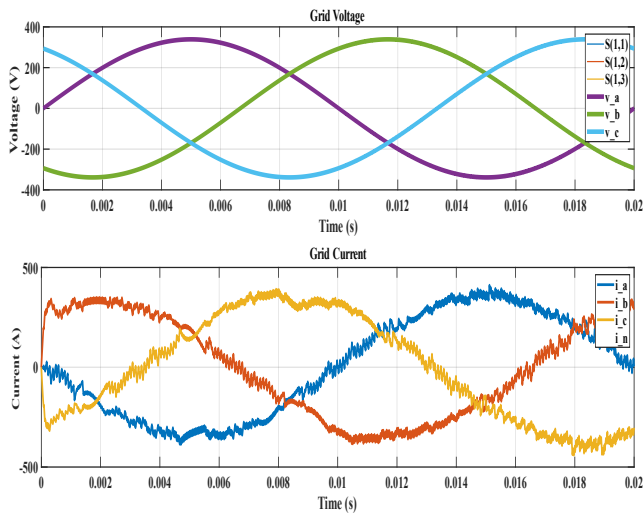


Fig. 6. EV charging station grid voltage and current.

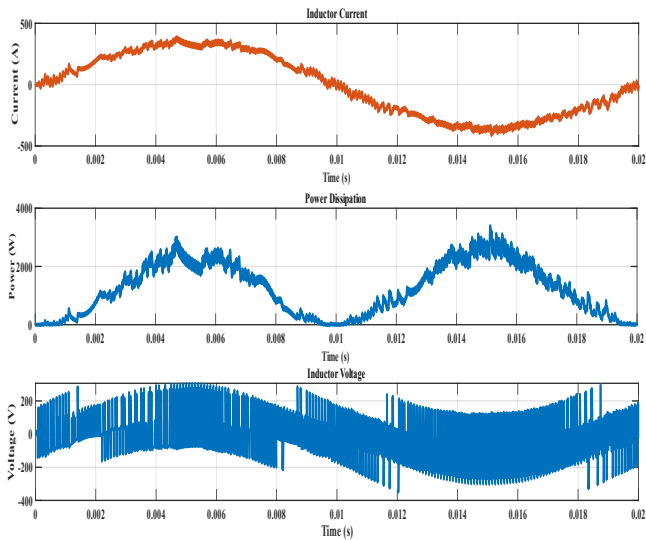


Fig. 7. EV charging station inductor current, power dissipation, and voltage.

The bottom part of Fig. 7 depicts the voltage of the inductor, characterized by a high-frequency ripple superimposed on a sinusoidal waveform. This high-frequency ripple is typical in power electronic circuits, resulting from the switching operations of the converter. The underlying sinusoidal component indicates the fundamental frequency of the system's operation. The voltage ripple's amplitude and frequency can provide insights into the converter's switching characteristics and efficiency. The steady current, combined with the varying power dissipation and voltage ripple, suggests that the system is dynamically balancing the load and maintaining overall performance.

Fig. 8 depicts the condition without any fault occurrence in the charger. Besides, Fig. 9 presents an analysis of DC voltage behavior in a battery and the occurrence of faults in an EV charger system over time. The top graph displays a steady increase in DC voltage, starting from a low value and rising linearly. This pattern indicates a consistent charging process, with the voltage climbing as the battery charges. The initial sharp rise in voltage suggests that the battery was initially at a low state of charge, and the charger quickly brought it up to a more optimal range.

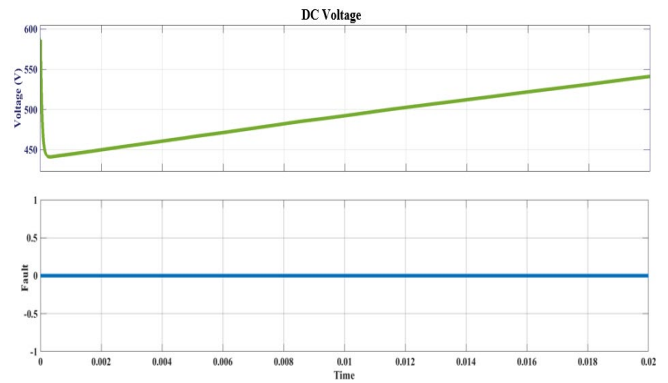


Fig. 8. Behavior of DC voltage for EV charging station without fault.

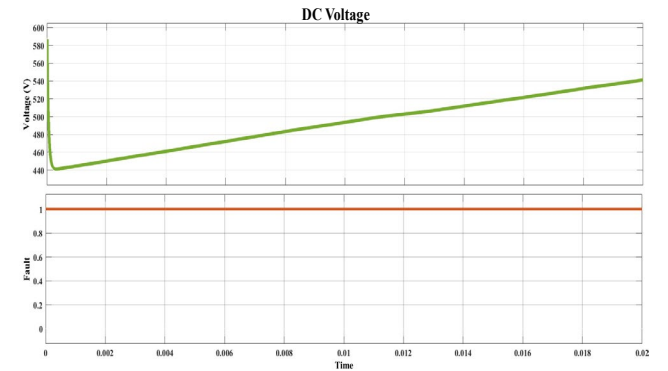


Fig. 9. Behavior of DC voltage for EV charging station with fault occurrence.

Following this, the voltage continues to rise steadily, reflecting a controlled and effective charging process characterized by a smooth voltage ramp-up without significant fluctuations. The bottom part of Fig. 9 highlights the occurrence of faults, which remain consistently low throughout the observed period. This indicates that the system operates without significant interruptions or malfunctions, demonstrating the reliability of the EV charger.

	0	1	10	11	100	101	110
0	3						
1		9					
10			9				
11				27			
100					9		
101						27	
110							27
	0	1	10	11	100	101	110

Fig. 10. Correlation matrix between predicted class and true class.

The correlation matrix provided in Fig. 10 represents the performance of a classification model in seven classes: 0, 1, 10, 11, 100, 101, and 110. Each cell in the matrix represents the number of instances predicted to be in a particular class, compared to the actual class. The diagonal cells, which are highlighted in varying shades of blue, indicate correct predictions: 3 for class 0, 9 for class 1, 9 for class 10, 27 for class 11, 9 for class 100, 27 for class 101, and 27 for class 110.

There are no off-diagonal elements, meaning there are no incorrect predictions, and all instances are classified correctly. The total sum of predictions is 111, leading to an accuracy of

around 94.59%. The varying shades of blue indicate the number of correct predictions, with darker shades representing higher counts. Fig. 11 graphically shows the developed algorithms fault detection accuracy for both training and validation. Initially, both the training and validation accuracies steadily increase and plateau around epoch 12. Despite some fluctuations, both curves maintain a relatively close alignment, indicating consistent performance between the training and validation sets. After epoch nearly 38, the curves continue to follow a similar trend, with both accuracies converging to a maximum value around 0.9459. This suggests that the model is well-trained and generalizes effectively to unseen data, as evidenced by the stable and high validation accuracy throughout most of the training process.

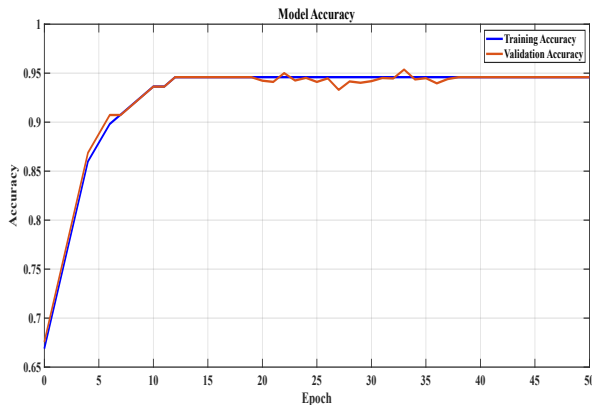


Fig. 11. Fault detection accuracy for training and validation.

V. CONCLUSION

The rapid shift towards Electric Vehicles (EVs) necessitates robust electrical system resilience, particularly focusing on the power electronic converters that not only drive the electric motors but also facilitate battery charging from the grid. This paper has presented a fault detection system for a standard EV charger to identify faults that could cause operational issues during the charging process. By implementing a Machine Learning (ML) classification algorithm, the system uses grid-side Total Harmonic Distortion (THD) and load resistance data to detect potential faults accurately. Extensive testing has shown that this method achieves an impressive fault prediction accuracy of 94.59%, highlighting its exceptional performance in terms of training and execution speed, as well as overall accuracy. This research demonstrates the significant benefits of applying advanced machine learning techniques to improve the reliability and efficiency of EV charging systems, thereby supporting a more seamless and accelerated transition to electric mobility.

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